Yide Lin

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Summary

Junior undergraduate seeking internship during the Spring/Summer of 2026. Relevant experience through coursework and personal projects with embedded systems/electronics; also programmed in C++ and Python. Highly results-driven with a strong ability to learn quickly, eager to apply knowledge to real-world products on the first day of the internship.

Education

Rensselaer Polytechnic Institute **Sep. 2023 – May 2027**
B.S., Electrical Engineering *Troy, NY*

- Rensselaer Leadership Award (2023–Present)
- Relevant Coursework:Embedded Control; Electric Circuits; Computer Component & Operation; Computer Science 1

Technical Skills

Languages : Python; C/C++; MATLAB
HDL : VHDL
Embedded : MSP432; Arduino; GPIO; Timers/Interrupts; PWM; I2C
Tools : Git/GitHub; LTspice; AutoCAD; Oscilloscope; Multimeter; Breadboarding & circuit debugging

Projects

TI-RSLK Differential-Drive Robot Firmware (MSP432, C) **April 2025**

- Developed MSP432 firmware in C using GPIO, timers, and interrupts; generated 25 kHz PWM for differential motor drive and direction control.
- Implemented a deterministic 50 ms control loop with quadrature encoder capture interrupts to estimate wheel speed from encoder timing; applied output clamping and integrator windup prevention for stability.
- Interfaced I^2C sensors to read IMU tilt (pitch/roll) and optional compass heading; mapped tilt angles to speed/turn commands and added PD heading correction to stabilize steering.

Smart Equipment for Vertical Hydroponic Farm – *IED Team Project* **May 2025 – August 2025**

- Designed and wired an Arduino Nano monitoring/control circuit using a DHT22 (temperature/humidity) and BH1750 (lux) sensor, and displayed live readings on an I^2C OLED.
- Implemented automatic lighting control: turned the grow light strip on/off based on ambient light with a 300 lux threshold; debugged sensor reads and display updates for stable operation.
- Contributed to circuit design, breadboard wiring, and firmware debugging; developed sensor polling and OLED UI logic and validated end-to-end behavior during system demo.

Indoor Pet Temperature Warning System – *Team Circuits Project* **May 2025 – August 2025**

- Collaborated in a team to design a low-cost temperature alert system for indoor pet enclosures using a Wheatstone bridge with an NTC thermistor and an OP484 op-amp comparator.
- Derived and tuned the threshold to trigger near 38°C; validated behavior in LTspice (25°C: Vin$\approx$2.02 V $<$ 2.5 V, 38°C: Vin$\approx$2.76 V $>$ 2.5 V) to confirm correct switching.
- Improved robustness by adding an RC de-glitch filter to suppress chatter near the threshold and a BJT driver stage to reliably actuate an LED and buzzer without loading the comparator output.

Interactive Sensor-Based Gift Box System – *Arduino, 3D Printing* **Spring 2026 – Present**

- Designed and prototyped an interactive gift box system integrating an HC-SR04 ultrasonic sensor, RGB LED, LCD1602 display, and Arduino Uno for lid-open detection and user feedback.
- Implemented embedded C/C++ logic to display custom messages and switch LED behavior from breathing-light mode to constant illumination based on measured distance changes.
- Created and iterated 3D-printed enclosure components, adjusting wall thickness, rounded corners, and sensor placement to improve fit, durability, and system reliability.
- Debugged hardware and firmware issues including unstable sensor readings, LED timing behavior, wiring layout, and component interaction during full-system testing.